ORVILLE ROCKWELL Jr.

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EDUCATION

Master's in Professional Studies – Applied Artificial Intelligence | Northeastern University *Expected 2027*

Research Interests: Machine Learning, Autonomous Systems & Computational Biology · Concentration: ML for Cybersecurity

Bachelor of Science – Information Technology | Purdue University Global *March 2026*

Chancellor's List 2026 · Asst. Dean's Honors Student Advisory Council · NSLS Honor Society

RESEARCH EXPERIENCE

Research Contributor – GPS-Denied Autonomous Navigation | Brown University, Prof. Benjamin Kimia *2026 – Pres.*

- Contributing to a research team developing GPS-denied drone localization, combining visual-inertial sensor fusion with RF-beacon triangulation to maintain positioning accuracy without satellite navigation.
- Own the multi-camera synchronization and calibration layer: brought up a synchronized camera array on a Raspberry Pi 5 with an Arducam Camarray HAT, contributing calibration routines and tests to the group's repository.

Team Lead – Volatility-Aware Distributional Forecasting (VAD-LSTM) | AAI 6640, Prof. Behnaz Merikhi *2026 – Pres.*

- Leading a four-person team building a recurrent network that forecasts a full probability distribution rather than a point estimate, extending the GARCH–LSTM equivalence of Zhao et al. (2024) with an in-cell variance recursion and a heavy-tailed Student-t output head.
- Own the ten-year point-in-time dataset; validate with purged walk-forward splits, isotonic recalibration and conformal coverage, benchmarked against market-implied prices.

Independent Research – Generative Drug Design for Post-Stroke Repair | AAI 6600 *2026*

- Designed a fully computational pipeline that proposes candidate compounds and scores each for blood-brain-barrier crossing, target activity and toxicity, using the combined score as a reinforcement-learning reward that retrains the generator.
- Applied scaffold-split validation and a held-out judging classifier to prevent circular scoring, reporting precision, recall and PR-AUC rather than accuracy alone.

Independent Research – Deep Learning Pipeline for Protein Stability & Drug-Binding Prediction *2026*

- Built a pipeline using pretrained protein language model embeddings (ESM) to predict how mutations affect protein stability and how strongly small molecules bind to target proteins, trained on public ChEMBL and ProTherm data.

Machine Learning for Market-Neutral Signal Discovery | AAI 6610 *2026*

- Pilot study testing whether learned signals survive risk adjustment, multiple-testing correction and realistic costs; purged, embargoed cross-validation with every trial logged and scored by a deflated Sharpe ratio to separate skill from selection bias.

WORK EXPERIENCE

Software Onboard Engineer | iTribe (B2C Blockchain & Fintech) *2023 – 2026*

- Built and shipped full-stack features across product rollouts spanning the Middle East, Africa, and North America.
- Led QA smoke and regression testing, sustaining a 90%+ release-readiness rate.

NOTABLE PROJECTS

Computer Vision Engineer – Marine Robotics | Northeastern University (NU Robotics) *2026 – Pres.*

- Developing and deploying computer vision models and sensor fusion for autonomous underwater vehicle perception as part of the software engineering division, contributing to entries in national robotics competitions.

CAD-LLM Platform | github.com/UriRock/CadLLM

- Converts natural-language specifications into structured CAD parameters through an LLM-backed REST pipeline; designed the prompt-to-parameter translation layer and evaluated outputs against manually specified designs.

APML – Analytical Predictive ML Platform | github.com/UriRock/Analytical-Predictive-ML-Platform

- End-to-end predictive pipeline from ingestion through feature engineering to training, benchmarking architectures in TensorFlow on both accuracy and inference latency.

TECHNICAL SKILLS

Languages: Python, C++, Rust, SQL, JavaScript, HTML, R, Ruby

ML/AI: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, OpenCV, ROS, Isaac Gym, MuJoCo, Stable-Baselines3

Tools/Infra: Docker, AWS, Linux, Git, REST APIs, Jira, Raspberry Pi, Wireshark